

ATOMiK Business Plan

ATOMiK · State-Aware Compute Architecture · Pre-Seed · May 2026

Company Overview

ATOMiK is a state-aware compute architecture startup developing a hardware/IP primitive for reducing wasted state movement where workload evidence supports it. The company is pre-revenue, pre-seed stage, hardware-validated on FPGA for selected proof paths, and raising \$2.0M to fund measured customer proof and IP diligence.

Entity status: current operating entity is Rockwell Industries, LLC; intended VC-ready Delaware C-Corp structure remains counsel-pending before institutional close. See data room for current legal-status documentation.

Problem

Many constrained systems - edge devices, embedded controllers, AI-at-the-edge systems, and remote industrial equipment - pay a hidden tax on state paths that repeatedly scan, compare, copy, move, sync, replay, or rebuild state even when only a compact change matters.

This waste manifests as:

- Battery pressure in edge and IoT devices
- Thermal pressure in sealed or fanless systems
- Bandwidth pressure over intermittent or expensive links
- Latency from rebuilding or replaying state that could have been tracked
- Hardware overbuild to handle peak-state throughput that is mostly redundant

Market scale: Data-center energy 415->945 TWh by 2030 (IEA). U.S. data centers 176->325-580 TWh by 2028 (LBNL). 66B liters direct cooling water (2023 U.S.). Semiconductor sales were \$791.7B in 2025 and Q1 2026 reached \$298.5B (SIA).

Sources: IEA Energy and AI executive summary (<https://www.iea.org/reports/energy-and-ai/executive-summary>); LBNL 2024 United States Data Center Energy Usage Report (<https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-states-data-center-energy-usage-report.pdf>); SIA 2025 global sales release (<https://www.semiconductors.org/global-annual-semiconductor-sales-increase-25-6-to-791-7-billion-in-2025/>); SIA Q1 2026 sales release (<https://www.semiconductors.org/global-semiconductor-sales-increase-25-from-q4-2025-to-q1-2026/>). These figures are market context only, not ATOMiK savings claims.

Solution: ATOMiK Architecture

Core equation: $\text{state} = \text{reference_state} \text{ XOR } \text{accumulated_delta}$

Operations: LOAD (set reference state) → ACCUM (accumulate deltas via XOR) → READ (reconstruct current state) → SWAP (commit epoch boundary)

The primitive is self-inverse, commutative, associative, and identity-preserving under XOR. Formal proof work exists in the repo, and selected algebraic checks have passed through the Linux userspace-to-FPGA validation path.

What this means for customers: evaluate whether a constrained workload can move, scan, sync, or rebuild less state while preserving correctness.

Current Proof Status

Hardware-validated proof exists for specific artifacts. The following is accurate as of May 2026:

- ATOMiK Desk v0.39-K: framebuffer-native prototype UI running on live Zynq FPGA (HARDWARE_VALIDATED)
- Algebraic tests: 16/16 PASS on XC7Z020 hardware through Linux userspace to FPGA (HARDWARE_VALIDATED)
- AX7020 performance matrix: raw board-run artifacts with 4-way comparison (LIVE_MEASURED with documented caveats)
- Formal proof foundation: FORMAL_PROOF where directly audited; otherwise proof work present. Avoid public proof counts until reconciled across repo, site, deck, and proof packet.
- SD boot artifacts: BOOT.bin and bitstream exist; standalone power-on boot remains gated (BUILD_ARTIFACT)

IMPORTANT: Battery, heat, cooling, water, and footprint outcomes require workload-specific customer measurement. These are evaluation targets, not current claims.

Target Market & Customer Segments

Segment	Pain & Target Metrics
Edge / embedded	Battery pressure, enclosure heat, intermittent links, reliability. Metrics: bytes moved, latency, bandwidth, power proxy.
AI at the edge	Context/state movement, memory pressure, response time. Metrics: context retained, transfer avoided.
Remote/industrial/robotics/defense	Weight, wattage, packet budget, field runtime. Metrics: runtime proxy, update cost.
Data center / infrastructure	Power bill, cooling, water pressure, rack density. Metrics: bytes moved, power/thermal path. (Strategic expansion)

Business Model

ATOMiK's near-term monetization is evaluation- and IP-led, not mass product manufacturing. Revenue path:

- Paid technical evaluations — bring one workload, one baseline, one constraint; receive measured fit assessment
- Design-partner engagements — deeper technical partnership with IP access
- IP licensing — architecture and implementation licensed to chip/platform companies

- Strategic optionality — licensing, integration partnership, or platform partnership if proof and IP diligence justify it

No revenue forecast until signed customer agreements. CFO approval required before projections are shared externally.

Go-to-Market

First sale: one constrained team with one state-heavy workload hitting a hard limit. Entry point: technical evaluation. Proof-gate: measured improvement on one agreed metric while preserving correctness.

Not: outspend incumbents. Not: build a consumer product. Not: chase the data-center market before edge proof exists.

Competitive Positioning

ATOMiK is not a CPU, GPU, NPU, or accelerator replacement. It is a state-aware architecture layer for workloads where state movement is the binding constraint.

Comparison	Differentiation
General CPUs	Flexible execution. ATOMiK differs by evaluating whether the workload can be represented as reference state plus accumulated change.
Compression	Reduces representation size. ATOMiK targets when state movement or repeated reconstruction can be avoided.
Caching	Keeps hot data nearby. ATOMiK targets meaningful change and epoch boundaries.
Custom accelerators	Optimize known kernels. ATOMiK is an architecture primitive that still requires workload-specific proof.

IP & Legal

Provisional patent filed (see data room: 03_intellectual_property). Patent conversion, prior-art review, trade-secret controls, and final IP assignment chain remain counsel-review workstreams. Pre-seed allocates \$300K to IP/legal.

Team

Matthew Rockwell — Founder. Hardware engineer, FPGA/ASIC development background. Built the ATOMiK architecture, validated on physical hardware. See data room for full founder profile.

Fractional CFO: needed before SAFE terms finalized. First technical hire and advisory board in progress.

Financial Summary

See ATOMiK_Financial_Model.xlsx for full detail. Summary:

Pre-seed target	\$2.0M (minimum \$1.25M, stretch \$2.75M)
Instrument	Planned SAFE — final terms require CFO/counsel
Runway	18 months at \$2.0M raise
Engineering	\$600K (demo hardening, measurement tooling)
Customer proof	\$400K (evaluation SOWs, baseline harnesses)
IP / legal	\$300K (patent conversion, diligence packet)
ASIC feasibility	\$300K (expert review, go/no-go)
Finance / GTM	\$250K + \$150K reserve
Tape-out	NOT funded in this round
Revenue	No forecast without signed agreements

Risks & Mitigations

Risk	Mitigation
Customer validation pending	Budget dedicated to proof reviews, technical evaluations, and design-partner outreach.
Battery/thermal/water outcomes unproven	Evaluation targets. Require workload measurement. Not claimed yet.
ASIC economics unvalidated	Funded in budget. Expert review before tape-out commitment.
IP protection coverage	Funded conversion. Prior-art review in progress.
Incumbent response	Build IP and proof quickly. Pursue partner conversations without implying a promised exit.